Quiz 02 – Make-Up

Emory University

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Instructions

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This quiz covers material from **Lectures 09–11**. Please write your answers in a Word document or PDF (preferred) and upload it to Canvas.

Guidelines:

- You are **encouraged** to use ChatGPT, Claude, Gemini, or other LLMs to help you think through these questions
- However, you must **write your answers in your own words** and demonstrate genuine understanding of the concepts
- Please **disclose which AI tool you used** (if any)
- Your answers should be thoughtful and show critical thinking, not just copy-pasted LLM outputs
- If applicable, please **attach screenshots** of your interactions with AI tools
- Submit your file via **Canvas** as a PDF (preferred) or Word document

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Question 1: Temperature for the job

Two users need AI for very different tasks:

- **User A** works at a law firm and needs an LLM to extract contract clauses *verbatim* from legal documents. Accuracy is everything; creative rewording could change the legal meaning.
- **User B** is a novelist who wants the LLM to brainstorm unexpected plot twists for a mystery story. She wants wild, surprising ideas.

Your task:

1. Explain what the **temperature** parameter does in an LLM. How does it affect the randomness of the model's output?
2. Recommend a temperature setting (low, e.g., 0.0–0.2, or high, e.g., 0.7–1.0) for each user and explain why.
3. What would go wrong if the settings were accidentally swapped – i.e., if the lawyer got high temperature and the novelist got low temperature?

Question 2: Jagged intelligence

An LLM correctly solves a complex calculus integral but then fails to count the number of "r"s in the word "strawberry." It confidently answers "2" (the correct answer is 3).

Your task:

1. Explain the concept of **jagged intelligence** (from Gans, 2024). What does it mean for LLMs to have "uneven" capabilities?
2. Why can a model handle advanced maths but stumble on a task a five-year-old could do? Think about how tokenisation and next-token prediction work.
3. What does jagged intelligence imply for how much you should trust an LLM's output, even when it sounds confident? Give a practical rule of thumb.

Question 3: The sycophancy trap

You tell an LLM: *"I think the Earth is about 6,000 years old. Am I right?"*

The LLM responds: *"That's an interesting perspective! There are indeed some traditions that support a young Earth timeline..."* – essentially agreeing with a scientifically incorrect claim.

Your task:

1. What is **sycophancy** in the context of LLMs, and why does it happen? (Think about how these models are trained to be "helpful.")
2. According to Geng et al. (2025), GPT changed its answers up to **55% of the time** after 10 rounds of user pushback. What does this tell you about how reliable an LLM's initial answer is when a user disagrees?
3. How does sycophancy contribute to **hallucinations**? Describe the chain: user asks → model doesn't know → model wants to be helpful → what happens next?

Question 4: Hallucination in the courtroom

In the 2023 case *Mata v. Avianca Airlines*, lawyer Steven Schwartz used ChatGPT for legal research. He submitted a court brief citing **six legal cases that did not exist**. ChatGPT had invented realistic-sounding case names, citations, and even quotations. When Schwartz asked ChatGPT *"Is Varghese a real case?"*, the model confirmed it was real. The judge sanctioned both attorneys.

Your task:

1. Why did ChatGPT invent plausible-sounding case names and citations? Explain using what you know about how LLMs generate text (next-token prediction, training data patterns).
2. Why did asking the model *"Is this real?"* not help? What does this tell you about an LLM's ability to verify its own outputs?
3. What should Schwartz have done differently? Describe at least two concrete verification steps he could have taken before submitting the brief.

Question 5: Keyword search vs semantic search

A company's HR manual says:

> *"Associates are entitled to 15 days of paid leave per calendar year."*

An employee asks the company's AI chatbot: *"How many vacation days do employees get?"*
Your task:

1. Explain why a **keyword search** system would fail to find the relevant passage. (Hint: compare the exact words in the query vs the document.)
2. Explain why a **semantic search** system would succeed. What technology makes this possible? (Use the concept of **embeddings** from Lecture 11.)
3. In a RAG (Retrieval-Augmented Generation) system, what happens *after* the relevant passage is retrieved? Describe the full pipeline: retrieval ⟶ what the LLM receives ⟶ what it generates.

Bonus question: RAG vs fine-tuning vs prompting

A hospital wants to deploy an AI assistant that answers staff questions about internal policies (infection control procedures, medication protocols, shift schedules, etc.). The policies are updated monthly. The hospital has limited budget and no in-house AI engineers.
Your task:

1. Should the hospital use **(a) prompt engineering**, **(b) RAG**, or **(c) fine-tuning**? Pick one and justify your choice using at least **three** of the following criteria: cost, setup time, data freshness, hallucination risk, ability to cite sources.
2. Briefly explain why each of the other two options would be a worse fit for this specific use case.
3. Even with the best approach, the system might still produce errors. Name one safeguard the hospital should put in place.